



NEWS81 ***December 2017***

<http://esanetworks.org/>

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Editorial

Dear *Flea News* Reader,

Though language translation of citations using computers is helpful, there are hazards. For example, the word 'flea' is sometimes translated by computers as 'chip'. It's unclear to me whether or not there are 'potato fleas' in other countries, or perhaps the word refers to the 'poker chips' used in card games. Magnetic stirring bars for beakers are also called 'fleas'.

In any case, the newsletter continues as "Flea News", and not "Chip News".

Yours in chips,
 R.L. Bossard, Editor, *Flea News*

Announcements

Dr Erica McAlister has recently taken over from Ms Theresa Howard as the curator in charge of the Collection at the Natural History Museum, London. They are preparing the collection for a digitisation project as well as a general upgrade to the facilities (those not within the Rothschild Cabinets).

**

This was the only flea paper presented at the 2017 Entomological Society of America meeting in Denver (November 5-8, 2017):

"Treating the Great gerbil (*Rhombomys opimus*) with a fipronil-based bait in an effort to control the vectors of cutaneous leishmaniasis (*Leishmania major*) and bubonic plague (*Yersinia pestis*) in southern Kazakhstan."

David Poche (davidp@genesislabs.com), Zaria Torres-Poche, Aidyn Eszhanov, Larisa Polyakova and Richard Poche, Genesis Laboratories, Inc, Wellington, CO, Kazakh Science Center for

the Quarantine of Zoonotic Diseases, Almaty, Kazakhstan.

**

There was a prairie dog die-off, due seemingly to plague, in Broomfield, Colorado in Mar 2017 near the Great Western Reservoir Open Space.

<http://www.sove.org/SOVE%20folder/Newsletter/SOVE%20Newsletter,%2006-2017.pdf>

**

“Is the number of flea bites per day known”?

In regards to the cat flea and its host cat or dog, is there any reference you may know on the feeding frequency of the female cat flea? Is there a difference between the ‘probing’ and the actual feeding?

Here I remember that cat flea bites on humans show more frequent probing without feeding, but with visible immune reactions, than actual blood meals. (Question from Norbert Mencke).

**

The 2018 Livestock Insect Workers Conference is to be held June 17-20 in San Juan, Puerto Rico. Contact information for the organizer and host, Manuel Lluberas, is Manuel@mosquitoden.com, while the direct LIWC contact is LIWC2018@gmail.com.

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Request for Rothschild Catalogue

Dear friends,

I need Volumes I, II, III, IV and V of "An Illustrated Catalogue of the Rothschild Collection of Fleas (Siphonaptera) in the British Museum (Natural History)", but I have not found anywhere.

If anyone can send me copies, or advise where to buy, I will be very happy.

Best regards...
ADEM KESKIN
ademkeskin@yahoo.com

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Total Number, and an Updated Listing, of Flea Species of the World
(continued)

Thanks to everyone sending corrections to the list.

R.L. Bossard
Editor, *Flea News*

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Hi everyone,

We are currently working on a project documenting the regionality of deer keds in the United States and we need your help! We would love to include any representative specimens of *Lipoptena cervi*, *L. mazamae*, *L. depressa*, or *Neolipoptena ferrisi* that you may have along with collection records. We would just need one representative sample per collection area (i.e., if you have multiple specimens of the same species from the same location, we only need a single specimen from that location). In terms of a "deadline," we are hoping to get records compiled by March 30th, so that is something to shoot for.

Specimens can be sent to: Michael Skvarla
501 ASI Building
University Park, PA 16802

If you have any questions, please feel free to contact Erika Machtinger (etm10@psu.edu) or Michael Skvarla (mxs1578@psu.edu). Thanks to everyone in advance. We greatly appreciate it!

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Dissertations

Bron, Gebbiena M.
Degree Ph.D.
Degree date 2017
Department of Comparative Biomedical Science
Subject Biology; Ecology; Veterinary services
Title
The Role of Short-lived Rodents and their Fleas in Plague Ecology on Prairie Dog Colonies
University location United States – Wisconsin

Multi-host and multi-vector diseases are complex. Plague, caused by the bacterium *Yersinia pestis*, affects many mammalian species and can be transmitted via several routes, but flea-borne transmission is the primary mode of transmission. Plague can decimate prairie dog (*Cynomys* spp.) colonies and other species that depend on them, including the endangered black-footed ferret (*Mustela nigripes*). After decades of research, plague dynamics on prairie dog colonies are still poorly understood. The purpose of this dissertation was to gain a holistic understanding of the complex dynamics of plague ecology in the western United States, by assessing the role of short-lived rodents and their fleas in plague maintenance and epizootics on prairie dog colonies.

Our study was conducted in conjunction with a large-scale, blind, placebo-controlled, field efficacy trial on the newly developed, orally-delivered sylvatic plague vaccine (SPV) for prairie dogs from 2013 to 2015, led by the U.S. Geological Survey, National Wildlife Health Center. Using a combination of field and laboratory-based studies, we observed that consumption of the SPV did not protect deer mice (*Peromyscus maniculatus*) and Northern grasshopper mice (*Onychomys leucogaster*) from plague and deer mouse abundance and short-lived rodent community structure (richness and evenness) were not significantly different between placebo and vaccine treated prairie dog colonies. Surprisingly, we detected plague positive mouse fleas (*Aetheca wagneri*, *Pleochaetis exilis*, *Orchopeas leucopus*) on mice prior to plague-induced declines in black-tailed prairie dogs (*C. ludovicianus*) in Montana and white-tailed prairie dogs (*C. leucurus*) in Wyoming, but not during plague outbreaks.

To determine the possible source of plague infection in the flea pools we used amplicon deep sequencing and we did not detect prairie dog DNA. In addition, rodent-flea interaction networks were highly specialized, geographically different, and fleas rarely switched from short-lived rodents to prairie dogs and vice versa. Thus we showed that mice are involved in the plague cycle on prairie dog colonies, but their role remains uncertain and the direct connection between short-lived rodents and prairie dogs stays anecdotal. Overall, these studies demonstrate the complexity involved in plague ecology and highlight the importance of a holistic approach to studying multi-host and multi-vector diseases.

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Beasley, E. M. (2017).
Island Biogeography of Small Mammals and Associated Ectoparasites in the Ozark Glades.
Missouri State Univ., 3203.
<https://bearworks.missouristate.edu/theses/3203>
Biology Masters Thesis

Island Biogeography Theory (IBT) explains and quantifies broad-scale ecological patterns among islands and isolated habitat patches. IBT predicts that the number of species per habitat patch varies as a function of area and isolation as a result of local colonization and extinction. IBT has been extended to habitat islands and hosts as islands for their associated parasites. In the latter system, host body mass acts as a substitute for island area, whereas host population density is used as a measure of isolation. Using the fragmented Ozark glades and the small mammals therein as a model system, I 1) used a multi-species occupancy model to correct for imperfect detection of species and generate species richnesses estimates, and 2) used the model to evaluate the effects of patch area, isolation, and shape on species richness. I then considered the mammalian hosts of the glades as islands for their associated ectoparasites, and evaluated 1) the effects of host population size on ectoparasite diversity metrics, and 2) tested for associations between mammalian host and ectoparasite communities. I captured a total of 8 mammal species and 7 ectoparasite orders in May–July 2016–2017. Glade area, but not isolation or shape, affected mammalian species richness. Ectoparasite richness and diversity were positively affected by host population size, and strong host-ectoparasite associations were present. IBT explains many of the patterns in small mammal and ectoparasite richness in the glade network, but matrix effects and host specificity may limit its applicability in this system.

**

Hund, Amanda Kathryn

Degree Ph.D.

Degree date 2017

Department of Ecology and Evolutionary Biology

Subject Biology; Evolution and Development; Parasitology

Title

How Parasite Communities Shape Host Evolution: From Individual-Level Experiments to Population-Level Comparisons

University location United States – Colorado

Parasites generate strong selective pressures that can shape host physiology, behavior, and evolutionary history. My dissertation research examines how complex interactions among hosts and parasites can shape patterns across spatial scales, from individual-level variation within populations to larger-scale patterns across populations, with emphasis on the association between parasites and evolution by mate-selection, or sexual selection. Sexual traits are used by females in mate-choice decisions to assess male quality and have often been linked to parasites within populations. Yet, broad patterns for links between sexual traits and parasites are unclear, and a unifying framework remains elusive. Thus, I first developed an approach for detecting connections between sexual traits, fitness costs, and the parasite community which I then applied in North American barn swallows, *Hirundo rustica erythrogaster*. I found that two costly parasites with different life histories were both associated with sexual trait expression. I performed further manipulation experiments with one of these parasites, nest mites, using an experimental technique that I developed. The results of these experiments show that fitness costs were due to size specific immune responses and mortality in nestlings and that a behavioral mechanism, territory settlement and defense, connected male sexual trait expression with parasites. Further, nest mites also influenced parental care in sex specific ways that were dynamic across development. To better understand how parasites may shape patterns of trait divergence among closely related populations, I tested whether adaptation to local parasite communities was tied to sexual trait divergence in the barn swallow species complex. I replicated my study of parasite communities and sexual traits in two additional [sub]species of barn swallow, *Hirundo rustica rustica* in the Czech Republic and *Hirundo rustica transitiva* in Israel. Within each population, sexual traits provide relevant information to females about local parasites, and associations between traits and parasites were flexible across populations, suggesting that divergent sexual traits are tied to variation in local ecology. Together, these results demonstrate that parasites interact to shape the behavior of their hosts in complex ways and that variation in parasites across closely related groups may play an important role in sexual trait divergence and speciation.

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Brook, Cara Elisabeth

Degree Ph.D.

Degree date 2017

Department of Ecology and Evolutionary Biology

Subject Ecology; Animal Diseases; Virology

Title

Elucidating Mechanisms of Viral Hosting in Bat Reservoirs for Emerging Zoonotic Disease

University location United States -- New Jersey

Bats are the acknowledged reservoir hosts for several of the world's most virulent emerging zoonotic diseases, including Hendra and Nipah henipaviruses, SARS and MERS coronaviruses, and Ebola and Marburg filoviruses. Bats appear to host these viruses—which are highly pathogenic in other mammals—while experiencing little to no ostensible morbidity or mortality, a phenomenon that has led researchers to question, increasingly, are bats somehow ‘special’ in their capacity as pathogen hosts?

Reviews of the literature indicate that bats, which comprise the second-most speciose mammalian order (Chiroptera) and make up ~20% of all mammalian diversity, host a disproportionate number of zoonotic viruses (those viruses which transmit from wildlife to human hosts) as compared with all other mammals.

This thesis explores the mechanisms by which bats host virulent infections, at both within-host and cross-population levels. In Chapter One, I delve deeply into the molecular biological underpinnings of bat metabolism, as relates to the evolution of flight, the one characteristic, which—above all—distinguishes Chiropterans from other mammals; I offer insights as to how such evolution might have shaped bat tolerance for, specifically, intracellular (often viral) infections. In Chapter Two, I explore the population-level dynamics of transmission for an intracellular bacterial infection in a community of sympatric Madagascar fruit bats, highlighting the critical reliance of these pathogens on species-specific host-parasite-vector relationships. In Chapter Three, I model the age-structured dynamics of transmission of henipaviruses and filoviruses infecting these same Madagascar fruit bats; I uncover a signature of elevated mortality in seropositive bats, a noteworthy indication of “disease” in these seemingly “special” viral hosts. In Chapter Four, I return once more to the molecular scale to model the dynamics of within-host viral transmission, using data describing the spread of viral infection across monolayers of bat cells. I highlight the critical reliance of bat cells on perpetually primed innate immune responses—a unique feature in mammalian immunity—to contain and control viral pathogenesis.

Bats are indeed special, it seems, in their capacity as viral hosts.

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Vaidya, Gaurav Girish

Degree Ph.D.

Degree date 2017

Department of Ecology and Evolutionary Biology

Subject Systematic biology

Title

Taxonomic Checklists as Biodiversity Data: How Series of Checklists can Provide Information on Synonymy, Circumscription Change and Taxonomic Discovery

University location United States – Colorado

Taxonomic checklists are a fundamental and widely-used product of taxonomy, providing a list of recognized taxa within a taxonomic group in a particular geographical area. Series of taxonomic

checklists provide snapshots of recognized taxa over a period of time. Identifying and classifying the changes between these checklists can provide information on rates of name, synonym and circumscription change and can improve aggregation of datasets reconciled to different checklists.

To demonstrate this, I used a series of North American bird checklists to test hypotheses about drivers of splitting rates in North America birds. In particular, I asked if splitting was predominantly undoing previous lumping that happened during the heyday of the modern synthesis.

I found that bird species have been split at an accelerating rate since the 1980s. While this was partially the result of previously lumped species being resplit, most splits were unrelated to previous lumps and thus represent new discoveries rather than simply the undoing of previous circumscription changes. I also used a series of North American freshwater algal checklists to measure stability over fifteen years, and found that 26% of species names were not shared or synonymized over this period. Rates of synonymization, lumping or splitting of species remained flat, a marked difference from North American birds. Species that were split or lumped (7% of species considered) had significantly higher abundance than other species in the USGS NAWQA dataset, a biodiversity database that uses these checklists as an index. They were associated with 19% of associated observations, showing that a small number of recircumscribed species could significantly affect interpretation of biodiversity data.

To facilitate this research, I developed a software tool that could identify and annotate taxonomic changes among a series of checklists, and could use this information to aggregate biodiversity data, which will hopefully facilitate similar research in the future. My dissertation demonstrates the value of taxonomic checklists series to answer specific questions about the drivers of taxonomic change ranging from philosophical and technical changes to characteristics of species themselves such as their abundance.

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Buhler, Andrea

Degree M.A.

Degree date 2017

Department of History

Fanciful but Not Forgotten: A Historical Examination of the Study of the Flea, 1840-1930

United States – Wisconsin

Although the Victorian and Progressive periods saw a rise in entomological research, fleas were not a priority for scientific investigation. The discovery of fleas as disease vectors in the late nineteenth century marked a turning-point in interest in fleas. Expanding this standard history of fleas, the thesis probes flea research conducted outside the confines of disease during 1840-1930. It documents and analyzes the contributions of Louis Bertolotto, William Heckler, Charles Rothschild, Karl Jordan, and L. O. Howard. Whereas those working in the new profession of entomology saw fleas as disease vectors, these men had different relationships with fleas: Bertolotto and Heckler engaged with fleas as entertainers, Rothschild and Jordan viewed fleas as collectible commodities to be catalogued, while Howard pursued fleas as household pests. Each relationship determined methods of study and questions asked. The thesis argues that the variant relationships provided impetus for substantial contributions to scientific understanding of fleas.

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Featured Research

Flea Taxonomy

Acosta, R. and M. W. Hastriter. 2017. "A Review of the Flea Genus *Phalacropsylla* Rothschild, 1915 (Siphonaptera, Ctenophthalmidae, Neopsyllinae, Phalacropsyllini) with New Host and Distributional Records." *ZooKeys* 2017 (675): 27-43. doi:10.3897/zookeys.675.12347.

Zurita, A., R. Callejón, M. De Rojas, and C. Cutillas. 2018. "Morphological and Molecular Study of the Genus *Nosopsyllus* (Siphonaptera: Ceratophyllidae). *Nosopsyllus barbarus* (Jordan & Rothschild 1912) as a Junior Synonym of *Nosopsyllus fasciatus* (Bosc d'Antic, 1800)." *Insect Systematics and Evolution* 49 (1): 81-101. doi:10.1163/1876312X-00002164.

In the present work, a comparative morphological and molecular study of *Nosopsyllus barbarus* (Jordan & Rothschild 1912) and *Nosopsyllus fasciatus* (Bosc, d'Antic 1800) isolated from rodents from different geographical regions (Spain, Morocco, Belgium, France and Portugal) has been carried out. The Internal Transcribed Spacers 1 and 2 (ITS1, ITS2) and partial 18S rRNA, and partial cytochrome c oxidase 1 (cox1) and cytochrome b (cytb) mtDNA sequences were determined to clarify the taxonomic status of these two species and to assess intraspecific variation and inter-specific sequence differences. In addition, a phylogenetic analysis with other species of fleas using Bayesian inference was performed. We have found morphological differences between *N. barbarus* and *N. fasciatus* that did not correspond with molecular differences. Furthermore, any of the five molecular markers used in this study was able to discriminate between the two species. Thus, based on the phylogenetic and molecular study of three nuclear markers (18S, ITS1, ITS2) and two mitochondrial markers (cox1 and cytb), as well as the concatenated dataset of both species, we concluded that morphological characters traditionally used to discriminate between the two species do not provide solid arguments for considering these two "morphospecies" as two different species. Therefore, we propose *N. barbarus* should be considered as a junior synonym of *N. fasciatus*.

[Editor's note: the authors fail to explain the consistent morphological differences between the two species' females.]

Zurita, A., R. Callejón, M. de Rojas, and C. Cutillas. 2017. "Morphological, Biometrical and Molecular Characterization of *Archaeopsylla erinacei* (Bouché, 1835)." *Bulletin of Entomological Research*: 1-13. doi:10.1017/S0007485317001274.

Hosts and Distribution

Lareschi, M., Cicuttin, G. L., De Salvo, M. N., Ibañez, L., & Montalti, D. (2017). The tropical fowl mite *Ornithonyssus bursa* (Acari: Mesostigmata: Macronyssidae) parasitizing the European starling *Sturnus vulgaris* (Aves: Passeriformes: Sturnidae), an invasive bird in central Argentina. An approach

to the bacterial fauna of this mite. *Revista Mexicana de Biodiversidad*, 88(2), 454-458.

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Ezquiaga, M. C., A. M. Abba, G. H. Cassini, and M. Lareschi. 2017. "Prevalence and Intensity of Fleas Parasitizing an Isolated Population of Screaming Hairy Armadillo in Buenos Aires Province, Argentina: Host-Related Factors and Temporal Dynamics." *Parasitology Research* 116 (11): 2895-2900. doi:10.1007/s00436-017-5595-7.

Fleas (Siphonaptera) of an isolated population of *Chaetophractus vellerosus* (Mammalia: Xenarthra) from Magdalena (Buenos Aires province) were studied, and their presence was associated with host-related factors (age, sex, weight, size, and physical condition) and temporal dynamics (seasonality and year). Three species of fleas were identified: *Polygenis (Polygenis) platensis* (Rhopalopsyllidae), *Tunga penetrans* (Tungidae), and *Pulex irritans* (Pulicidae). Prevalences were significant for year, season, and physical condition. Intensities were significantly different for year, physical condition, and weight. The intensities of fleas were higher in 2009 than in other years, probably because of lower rainfall than the annual average leading to extremely dry climatic conditions in 2008. Intensities decreased in individuals with major body weight and increased in individuals with poor physical condition. In this study, the dynamics of the flea community associated with an armadillo population is analyzed for the first time taking into account host-related factors and temporal dynamics, and also how these factors influence the community.

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Lareschi M., Procopio D.E., Frere E., Morgenthaler A., Millones A. & Barrionuevo M. Nuevo registro de pulgas (Hexapoda, Siphonaptera) parasitando al Pingüino de Magallanes *Spheniscus magellanicus* (Forster) (Aves, Sphenisciformes, Spheniscidae) en la Isla Quiroga, Puerto Deseado, Santa Cruz, Argentina. *Revista Argentina de Parasitología* 9-RevArgParasitol Vol 5.

Fleas from the genus *Parapsyllus* Enderlein, 1903 (Rhopalopsyllidae, Parapsyllinae) exclusively parasitize birds, mainly marine species. We present a new record of *Parapsyllus longicornis* (Enderlein, 1901) parasitizing the Magellanic Penguin *Spheniscus magellanicus* (Forster) (Aves, Sphenisciformes, Spheniscidae) in Argentina. This new record extends the northern limit of distribution of this flea in the Americas.

**

Kwak, M.L., Madden, C., Wicker, L. (2017)

The first record of the native flea *Acanthopsylla* Rainbow, 1905 (Siphonaptera: Pygiopsyllidae) from the endangered Tasmanian devil (*Sarcophilus harrisii*, 1841), with a review of the fleas associated with the Tasmanian devil.

Australian Entomologist, 44, pp. 293-296.

**

Jensen, J.-K., Magnussen, E. (2017)

Flea species (Insecta, Siphonaptera) from the Faroe Islands.

Norwegian Journal of Entomology, 64 (1), pp. 22-27.

The current investigation specifies the occurrence of four flea species living in the Faroe Islands: *Ceratophyllus garei* (Rothschild, 1902), *Dasypsyllus gallinulae gallinulae* (Dale, 1878), *Leptopsylla segnis* (Schönherr, 1811) and *Ctenophthalmus nobilis* (Rothschild, 1902). One of these species, *C. garei*, has never been reported in the Faroe Islands before. In addition, the flea species that have been reported in the Faroe Islands during the years 1929 to 2016 are listed. In total eleven species have been found. However, one of them, *Pulex irritans* (Linnaeus, 1758), has probably disappeared again.

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SANCHEZ J.P., EZQUIAGA, M.C. & M. RUIZ. Fleas (Insecta: Siphonaptera) with public health relevance in domestic pigs (Artiodactyla: Suidae) from Argentina. Zootaxa (In press)

SANCHEZ J.P., TRAVAINI, A., RODRÍGUEZ, A. & M. LARESCHI. PRIMER REGISTRO DE PULGAS (INSECTA, SIPHONAPTERA) EN POBLACIONES DE *Lycalopex* (CARNIVORA, CANIDAE) DE LA PATAGONIA ARGENTINA. MASTOZOOLOGÍA NEOTROPICAL (In press)

SANCHEZ J.P. & Y. GUROVICH. FLEAS (INSECTA: SIPHONAPTERA) ASSOCIATED TO THE ENDANGERED NEOTROPICAL MARSUPIAL MONITO DEL MONTE (*Dromiciops gliroides*) MICROBIOTHERIA: MICROBIOTHERIIDAE) IN ITS SOUTHERNMOST POPULATION OF ARGENTINA. MASTOZOOLOGÍA NEOTROPICAL (In press)

Ecology

Rakotomanga, M. N., Goodman, S. M., Soarimalala, V., Boyer, S. & Apanaskevich, D. 2017. Les tiques dures (Acari : Ixodidae) ectoparasites de micromammifères non-volants dans la forêt d'Ambohitantely, Madagascar. Malagasy Nature 12: 59-67.

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Randrenjarison Andriniaina HR, Beaucournu AD, Soarimalala V., Boyer S., Goodman SM. 2017. Sex ratio in fleas (Insecta: Siphonaptera) of Ambohitantely, Central Highlands of Madagascar. Malagasy Nature, 12: 42-50.

The sex ratio of Siphonaptera (fleas), ectoparasites of small endemic (Afrosoricida: Tenrecidae, and Nesomyidae: Nesomyinae) and introduced (Soricidae and Muridae) mammals, was investigated based on field collections made at the end of the wet season (31 March to 9 April 2014) and the dry season (15 to 25 October 2014) at a site in the Central Highlands of Madagascar (Réserve Spéciale d'Ambohitantely). The study site, located in central Madagascar, is 140 km northwest of Antananarivo along Route National No.4 (towards Mahajanga). Ambohitantely is located on the

Ankazobe Tampoketsa (plateau) between longitude 47°12' and 47°20'E, and latitude 18°04' and 19°14'S, and the elevational range falls between 1550 and 1662 m above sea-level. Small mammals were captured during the two sessions, each 10 days, using the same protocol: 1) two styles of live traps, Sherman and National (baited with peanut butter), in a ratio of 1:4, and approximately 70% placed on the ground and the balance in arboreal sets; these traps were intended to capture Nesomyinae and Muridae rodents and to a lesser extent Tenrecidae; and 2) three lines of pit-fall traps employed to trap Tenrecidae and to a lesser extent rodents. This latter type of trap was composed of 11 plastic buckets, each with a capacity of 15 l, placed 10 m apart, buried in the earth so the upper rim was flush with the ground surface, and with a 100 m long and 0.75 m high plastic drift fence stapled to vertical posts and bisecting each bucket. During the two field surveys, 289 individual small mammals were captured, including 12 endemic and four introduced species. Each of these individuals was inspected for ectoparasites, yielding a total of 339 flea specimens, composed of two genera and nine species (*Synopsyllus estradei*, *S. fonquerniei*, *Paractenopsyllus duplantieri*, *P. grandidieri*, *P. kerguisteli*, *P. petiti*, *P. rouxi*, *P. vauceli*, and *P. viettei*). All of these flea species are endemic to Madagascar and, surprisingly given the number of non-native small mammals captured, no introduced flea was collected. As the trapping and handling techniques between the two surveys were in parallel, it is possible to compare seasonal differences in aspects of the ecology of the collected fleas and their hosts, including sex-ratios. The results for the nine species of identified fleas indicate that the sex-ratio was not equal for males and females and in most cases in favor of females; this is similar to the pattern found in different species of ectoparasites, including fleas. In *S. estradei*, female fleas outnumbered males and this result was statistically significant (Wilcoxon, $W = 224$, $P = 0.027$). Some seasonal variation in the sex ratio of fleas was also observed. In addition, certain flea species show apparent preferences to the sex of the host, as well as showing seasonal variation.

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Santos, Ana M. C.; Field, Richard; Ricklefs, Robert E. 2016

New directions in island biogeography.

GLOBAL ECOLOGY AND BIOGEOGRAPHY Volume: 25 Issue: 7 Pages: 751-768

[Editor's note: the usefulness of the island biogeography paradigm is reviewed, and mentions the host-as-island hypothesis in connection to parasites, including fleas.]

**

Eads, D.A., Hoogland, J.L. (2017)

Precipitation, Climate Change, and Parasitism of Prairie Dogs by Fleas that Transmit Plague.

Journal of Parasitology, 103 (4), pp. 309-319.

Fleas (Insecta: Siphonaptera) are hematophagous ectoparasites that can reduce the fitness of vertebrate hosts. Laboratory populations of fleas decline under dry conditions, implying that populations of fleas will also decline when precipitation is scarce under natural conditions. If precipitation and hence vegetative production are reduced, however, then herbivorous hosts might

suffer declines in body condition and have weakened defenses against fleas, so that fleas will increase in abundance. We tested these competing hypotheses using information from 23 yr of research on 3 species of colonial prairie dogs in the western United States: Gunnison's prairie dog (*Cynomys gunnisoni*, 1989-1994), Utah prairie dog (*Cynomys parvidens*, 1996-2005), and white-tailed prairie dog (*Cynomys leucurus*, 2006-2012). For all 3 species, flea-counts per individual varied inversely with the number of days in the prior growing season with >10 mm of precipitation, an index of the number of precipitation events that might have caused a substantial, prolonged increase in soil moisture and vegetative production. Flea-counts per Utah prairie dog also varied inversely with cumulative precipitation of the prior growing season. Furthermore, flea-counts per Gunnison's and white-tailed prairie dog varied inversely with cumulative precipitation of the just-completed January and February. These results complement research on black-tailed prairie dog (*Cynomys ludovicianus*) and might have important ramifications for plague, a bacterial disease transmitted by fleas that devastates populations of prairie dogs. In particular, our results might help to explain why, at some colonies, epizootics of plague, which can kill >95% of prairie dogs, are more likely to occur during or shortly after periods of reduced precipitation. Climate change is projected to increase the frequency of droughts in the grasslands of western North America. If so, then climate change might affect the occurrence of plague epizootics among prairie dogs and other mammalian species that associate with them.

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Eads, D.A., Biggins, D.E. (2017)

Paltry past-precipitation: Predisposing prairie dogs to plague?

Journal of Wildlife Management, 81 (6), pp. 990-998.

Evolution

Medvedev, S.G. (2017)

Adaptations of fleas (Siphonaptera) to parasitism.

Entomological Review, 97 (8), pp. 1023-1030.

The paper deals with peculiarities of flea structure determined by their parasitism on mammals and birds. On the basis of the data on diversity of morphological characters, the leading role of structures of the frontal and nototrochanteral complexes in the adaptive evolution of Siphonaptera is substantiated. Peculiarities of the pulicoid, ischnopsylloid, palaeopsylloid, and generalized morphological types are analyzed together with examples of narrow morphological specializations. Distribution of fleas of these morphological types over five groups of hosts differing in the degree of mobility and association with nests and burrows is also analyzed.

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Harimalala, M., Telfer, S., Delatte, H., Watts, P.C., Miarinjara, A., Ramihangihajason, T.R., Rahelinirina, S., Rajerison, M., Boyer, S. (2017)

Genetic structure and gene flow of the flea *Xenopsylla cheopis* in Madagascar and Mayotte.

Parasites and Vectors, 10 (1), art. no. 347 (unpaginated).

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Pathology and Control

Dryden, M. W., M. S. Canfield, E. Niedfeldt, A. Kinnon, K. Kalosy, A. Smith, K. M. Foley, et al. 2017. "Evaluation of Sarolaner and Spinosad Oral Treatments to Eliminate Fleas, Reduce Dermatologic Lesions and Minimize Pruritus in Naturally Infested Dogs in West Central Florida, USA." *Parasites and Vectors* 10 (1). doi:10.1186/s13071-017-2328-9.

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Rust, Michael K. 2017. The Biology and Ecology of Cat Fleas and Advancements in Their Pest Management: A Review. *INSECTS* Volume: 8 Issue: 4 Article Number: 118.

The cat flea *Ctenocephalides felis felis* (Bouche) is the most important ectoparasite of domestic cats and dogs worldwide. It has been two decades since the last comprehensive review concerning the biology and ecology of *C. f. felis* and its management. Since then there have been major advances in our understanding of the diseases associated with *C. f. felis* and their implications for humans and their pets. Two rickettsial diseases, flea-borne spotted fever and murine typhus, have been identified in domestic animal populations and cat fleas. Cat fleas are the primary vector of *Bartonella henselae* (cat scratch fever) with the spread of the bacteria when flea feces are scratched in to bites or wounds. Flea allergic dermatitis (FAD) common in dogs and cats has been successfully treated and tapeworm infestations prevented with a number of new products being used to control fleas. There has been a continuous development of new products with novel chemistries that have focused on increased convenience and the control of fleas and other arthropod ectoparasites. The possibility of feral animals serving as potential reservoirs for flea infestations has taken on additional importance because of the lack of effective environmental controls in recent years. Physiological insecticide resistance in *C. f. felis* continues to be of concern, especially because pyrethroid resistance now appears to be more widespread. In spite of their broad use since 1994, there is little evidence that resistance has developed to many of the on-animal or oral treatments such as fipronil, imidacloprid or lufenuron. Reports of the perceived lack of performance of some of the new on-animal therapies have been attributed to compliance issues and their misuse. Consequentially, there is a continuing need for consumer awareness of products registered for cats and dogs and their safety.

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Marcondes, C. B. (2017). Arboviruses: a new and useful term for diseases related to arthropods. *Revista da Sociedade Brasileira de Medicina Tropical*, 50(3), 433-433.

During the Cretaceous and Jurassic eras (145-65 MYA), blood feeding evolved at least six times in approximately 15,000 arthropod species in 400 genera. Four orders of insects include blood-feeders: Pthiraptera [*sic*, Phthiraptera], Siphonaptera, Hemiptera, and Diptera. Some moths (Noctuidae) also feed on blood, and some chewing lice (Mallophaga) and flies with rasping structures in the tips of their

probosces [*sic*, proboscises or proboscides] will possibly evolve to full blood feeding. All ticks (ca. 870 species in Acari: Ixodida) and several mites suck blood, and several ticks and mites may respectively cause paralysis and mange. Fly larvae can develop in or on the bodies of several animals. Several insects can transport microorganisms on their legs and in their intestinal tracts, which are then mechanically transmitted to other organisms. Some Arachnida (centipedes, spiders, scorpions, etc.) and hymenopteran (bees, wasps, and ants), heteropteran (true bugs) and other insects may cause envenomation.

By sucking blood, arthropods can cause direct damage, including allergy [editor's note: allergy might be thought of as indirect damage due to the mediation of the host's immune reaction], and transmit many potentially pathogenic microorganisms, including more than 500 arboviruses, bacteria, protozoa, and helminthes. Although the modes of transmission are diverse, these potential pathogens have the common character of “hitchhiking” on/in arthropods, which protects them from environmental aggressions and allows them to be transported and distributed in efficient ways.

Diseases related to arthropods have been grouped as diseases transmitted and caused by arthropods, always needing the additional explanation that, in reality, arthropods transmit potential pathogens, and not diseases, which are a possible consequence of infection. Other expressions are arthropod-borne diseases and arthropod borne diseases [*sic*], which are more generic and require no additional explanation.

Viruses transmitted by arthropods were initially called the mosquito-borne viral encephalitides, which was later changed to arthropod-borne virus encephalitides (when the involvement of ticks and *Culicoides* in several cycles became obvious) and arborviruses; finally being corrected to arboviruses, to prevent any association to trees (arbor in Latin). Similarly, the acronym arboterrorism was recently proposed to terrorism caused by arthropods.

Because the utilization of arthropod borne diseases to represent all diseases in some way related to arthropods presents quite a mouthful and is somewhat cumbersome, I propose the adoption of arbodiseases, instead. The utilization of this word would facilitate communication and emphasize the relationship between these diseases and the need for preventive measures, besides stimulating comparative studies.

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Ditrich, H. (2017)

The transmission of the Black Death to Western Europe: a critical review of the existing evidence.

Mediterranean Historical Review, 32 (1), pp. 25-39.

In 1347, the Black Death was introduced from the north-eastern coast of the Black Sea and the Sea of Asov towards southern and western Europe, where it then spread dramatically. A report by the Italian chronicler Gabriel de Mussis of the siege of Caffa (1345–47) is often credited as describing an early deployment of a “biological weapon”, thus triggering the “Black Death” in western Europe. He reports that Mongol troops threw plague victims into the city with catapults, thus contaminating the

inhabitants. However, re-evaluation of historical, biological and epidemiological data indicates that the spread of the disease was probably an inevitable consequence of the intense trade relations along the coasts of the Black Sea and the Mediterranean. Therefore, the alleged catapulting of infected corpses would rather have been a marginal contribution to the diffusion of the disease (if it took place at all). The infection was subsequently spread by refugee ships via ports at Constantinople and along the Mediterranean trading routes and harbours towards Genoa, Marseille and Venice, thus initiating the Plague in Europe. The further propagation of the disease inland is still a matter of controversial discussions. However, epidemiological data indicate that the most essential animal vector for further distribution of the plague in central and northern Europe was probably the human louse (*Pediculus humanis* [*sic, humanus*]), instead of the oriental (or tropical) rat flea (*Xenopsylla cheopis*).

[Editor's note: the louse? See next citation.]

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Katharine R. Dean, Fabienne Krauer, Lars Walløe, Ole Christian Lingjærde, Barbara Bramanti, Nils Chr. Stenseth, Boris V. Schmid. 2018.

Human ectoparasites and spread of plague in Europe

Proceedings of the National Academy of Sciences Jan 2018, 201715640; DOI: 10.1073/pnas.1715640115

Plague, caused by the bacterium *Yersinia pestis*, can spread through human populations by multiple transmission pathways. Today, most human plague cases are bubonic, caused by spillover of infected fleas from rodent epizootics, or pneumonic, caused by inhalation of infectious droplets. However, little is known about the historical spread of plague in Europe during the Second Pandemic (14–19th centuries), including the Black Death, which led to high mortality and recurrent epidemics for hundreds of years. Several studies have suggested that human ectoparasite vectors, such as human fleas (*Pulex irritans*) or body lice (*Pediculus humanus humanus*), caused the rapidly spreading epidemics. Here, we describe a compartmental model for plague transmission by a human ectoparasite vector. Using Bayesian inference, we found that this model fits mortality curves from nine outbreaks in Europe better than models for pneumonic or rodent transmission. Our results support that human ectoparasites were primary vectors for plague during the Second Pandemic, including the Black Death (1346–1353), ultimately challenging the assumption that plague in Europe was predominantly spread by rats.

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Andrades Valtueña, A., Mitnik, A., Key, F.M., Haak, W., Allmäe, R., Belinskij, A., Daubaras, M., Feldman, M., Jankauskas, R., Janković, I., Massy, K., Novak, M., Pfrengle, S., Reinhold, S., Šlaus, M., Spyrou, M.A., Szécsényi-Nagy, A., Törv, M., Hansen, S., Bos, K.I., Stockhammer, P.W., Herbig, A., Krause, J.

The Stone Age Plague and Its Persistence in Eurasia.

(2017) *Current Biology*, 27 (23), pp. 3683-3691.e8.

Yersinia pestis, the etiologic agent of plague, is a bacterium associated with wild rodents and their fleas. Historically, it was responsible for three pandemics: the Plague of Justinian in the 6th century AD, which persisted until the 8th century; the renowned Black Death of the 14th century, with recurrent outbreaks until the 18th century; and the most recent 19th century pandemic, in which *Y. pestis* spread worldwide and became endemic in several regions. The discovery of molecular signatures of *Y. pestis* in prehistoric Eurasian individuals and two genomes from Southern Siberia suggest that *Y. pestis* caused some form of disease in humans prior to the first historically documented pandemic. Here, we present six new European *Y. pestis* genomes spanning the Late Neolithic to the Bronze Age (LNBA; 4,800 to 3,700 calibrated years before present). This time period is characterized by major transformative cultural and social changes that led to cross-European networks of contact and exchange. We show that all known LNBA strains form a single putatively extinct clade in the *Y. pestis* phylogeny. Interpreting our data within the context of recent ancient human genomic evidence that suggests an increase in human mobility during the LNBA, we propose a possible scenario for the early spread of *Y. pestis*: the pathogen may have entered Europe from Central Eurasia following an expansion of people from the steppe, persisted within Europe until the mid-Bronze Age, and moved back toward Central Eurasia in parallel with human populations. Andrades Valtueña, et al. present the first six European *Y. pestis* genomes dating from the Late Neolithic and the Early Bronze Age. These data suggest that *Y. pestis* entered Europe during a human migration around 4800 BP, persisted in Europe, and traveled back to Central Eurasia.

Microbial Symbiotes

Healy, S.P., Brown, L.D., Hagstrom, M.R., Foil, L.D., Macaluso, K.R. (2017)

Effect of *Rickettsia felis* strain variation on infection, transmission, and fitness in the cat flea (Siphonaptera: Pulicidae).

Journal of Medical Entomology, 54 (4), pp. 1037-1043.

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Llanos, S., Suazo, C.G., Quillfeldt, P., Cursach, J.A., Salas, L.M. (2018)

Ectoparasite abundance and apparent absence of hemoparasites in two albatross species in Sub-Antarctic Chile.

Polar Biology, 41 (1), pp. 143-147.

The Black-browed Albatross (BBA, *Thalassarche melanophrys*) and the Grey-headed Albatross (GHA, *Thalassarche chrysostoma*) are long-lived sympatric species breeding in the Atlantic and Pacific sections of the Southern Ocean. While the Atlantic populations of these two species have been the

subject of several studies, little information is available on the most important colonies in the Southeast Pacific Ocean (Chile). The presence of parasites in these long-lived hosts has been described for the colonies in the Southern Atlantic, but not in those from Chile. Therefore, the objective of this study was to identify ectoparasites and hemoparasites associated with both species breeding in the Diego Ramírez Archipelago (56°S) in Southern Chile. Chicks were examined for ectoparasites during the breeding seasons in 2010 and 2011 (GHA, n = 80; BBA, n = 40). Parasites were collected manually, mounted, and identified to species level. The sex and developmental stage of adult ectoparasites were also determined. Ectoparasites were detected on 78.75 and 32.50% of GHA and BBA chicks, respectively. A total of 126 ectoparasites were obtained: 103 *Ixodes uriae* ticks (GHA 79.62%, BBA 20.38%); 22 *Parapsyllus longicornis* fleas (GHA = 31.8%, BBA = 68.2%); and a single *Austromenopon affine* louse (found on GHA). Diego Ramírez represents a new record and distribution range for *I. uriae* and *A. affine*. The presence of hemoparasites was explored on blood smears for GHA (n = 26) and BBA (n = 119), but none were recorded in either of these albatrosses.

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Fleas in Art and History

The Decameron of Giovanni Boccaccio (excerpt)

Author: Giovanni Boccaccio (Certaldo, Republic of Florence, 16 June 1313 – 21 December 1375)

Translator: John Payne.

Publisher: The Villon Society, London, 1886, Walter J. Black, Inc. New York.

As often, most gracious ladies, as, taking thought in myself, I mind me how very pitiful you are all by nature, so often do I recognize that this present work will, to your thinking, have a grievous and a weariful beginning, inasmuch as the dolorous remembrance of the late pestiferous mortality, which it beareth on its forefront, is universally irksome to all who saw or otherwise knew it. But I would not therefore have this affright you from reading further, as if in the reading you were still to fare among sighs and tears. Let this grisly beginning be none other to you than is to wayfarers a rugged and steep mountain, beyond which is situate a most fair and delightful plain, which latter cometh so much the pleasanter to them as the greater was the hardship of the ascent and the descent; for, like as dolour occupieth the extreme of gladness, even so are miseries determined by imminent joyance. This brief annoy (I say brief, inasmuch as it is contained in few pages) is straightway succeeded by the pleasance and delight which I have already promised you and which, belike, were it not aforesaid, might not be looked for from such a beginning. And in truth, could I fairly have availed to bring you to my desire otherwise than by so rugged a path as this will be I had gladly done it; but being in a manner constrained thereto, for that, without this reminiscence of our past miseries, it might not be shown what was the occasion of the coming about of the things that will hereafter be read, I have brought myself to write them.

I say, then, that the years [of the era] of the fruitful Incarnation of the Son of God had attained to

the number of one thousand three hundred and forty-eight, when into the notable city of Florence, fair over every other of Italy, there came the death-dealing pestilence, which, through the operation of the heavenly bodies or of our own iniquitous dealings, being sent down upon mankind for our correction by the just wrath of God, had some years before appeared in the parts of the East and after having bereft these latter of an innumerable number of inhabitants, extending without cease from one place to another, had now unhappily spread towards the West. And thereagainst no wisdom availing nor human foresight (whereby the city was purged of many impurities by officers deputed to that end and it was forbidden unto any sick person to enter therein and many were the counsels given for the preservation of health) nor yet humble supplications, not once but many times both in ordered processions and on other wise made unto God by devout persons,—about the coming in of the Spring of the aforesaid year, it began on horrible and miraculous wise to show forth its dolorous effects. Yet not as it had done in the East, where, if any bled at the nose, it was a manifest sign of inevitable death; nay, but in men and women alike there appeared, at the beginning of the malady, certain swellings, either on the groin or under the armpits, whereof some waxed of the bigness of a common apple, others like unto an egg, some more and some less, and these the vulgar named plague-boils. From these two parts the aforesaid death-bearing plague-boils proceeded, in brief space, to appear and come indifferently in every part of the body; wherefrom, after awhile, the fashion of the contagion began to change into black or livid blotches, which showed themselves in many on the arms and about the thighs and every other part of the person, in some large and sparse and in others small and thick-sown; and like as the plague-boils had been first (and yet were) a very certain token of coming death, even so were these for every one to whom they came.

To the cure of these maladies nor counsel of physician nor virtue of any medicine appeared to avail or profit aught; on the contrary,—whether it was that the nature of the infection suffered it not or that the ignorance of the physicians (of whom, over and above the men of art, the number, both men and women, who had never had any teaching of medicine, was become exceeding great,) availed not to know whence it arose and consequently took not due measures thereagainst,—not only did few recover thereof, but well nigh all died within the third day from the appearance of the aforesaid signs, this sooner and that later, and for the most part without fever or other accident. And this pestilence was the more virulent for that, by communication with those who were sick thereof, it gat hold upon the sound, no otherwise than fire upon things dry or greasy, whenas they are brought very near thereunto. Nay, the mischief was yet greater; for that not only did converse and consortion with the sick give to the sound infection of cause of common death, but the mere touching of the clothes or of whatsoever other thing had been touched or used of the sick appeared of itself to communicate the malady to the touch. A marvelous thing to hear is that which I have to tell and one which, had it not been seen of many men's eyes and of mine own, I had scarce dared credit, much less set down in writing, though I had heard it from one worthy of belief. I say, then, that of such efficiency was the nature of the pestilence in question in communicating itself from one to another, that, not only did it pass from man to man, but this, which is much more, it many times visibly did;—to wit, a thing which had pertained to a man sick or dead of the aforesaid sickness, being touched by an animal foreign to the human species, not only infected this latter with the plague, but in a very brief space of time killed it. Of this mine own eyes (as hath a little before been said) had one day, among others, experience on this wise; to wit, that the rags of a poor man, who had died of the plague, being cast out into the public way, two hogs came up to them and having first, after their wont, rooted among them with their snouts, took them in their mouths and tossed them about their jaws; then, in a little while, after turning round and round, they both, as if they had taken poison, fell down dead upon the rags with which they had in an ill hour intermeddled.

From these things and many others like unto them or yet stranger divers fears and conceits were begotten in those who abode alive, which well nigh all tended to a very barbarous conclusion, namely,

to shun and flee from the sick and all that pertained to them, and thus doing, each thought to secure immunity for himself. Some there were who conceived that to live moderately and keep oneself from all excess was the best defence against such a danger; wherefore, making up their company, they lived removed from every other and shut themselves up in those houses where none had been sick and where living was best; and there, using very temperately of the most delicate viands and the finest wines and eschewing all incontinence, they abode with music and such other diversions as they might have, never suffering themselves to speak with any nor choosing to hear any news from without of death or sick folk. Others, inclining to the contrary opinion, maintained that to carouse and make merry and go about singing and frolicking and satisfy the appetite in everything possible and laugh and scoff at whatsoever befell was a very certain remedy for such an ill. That which they said they put in practice as best they might, going about day and night, now to this tavern, now to that, drinking without stint or measure; and on this wise they did yet more freely in other folk's houses, so but they scented there aught that liked or tempted them, as they might lightly do, for that every one—as he were to live no longer—had abandoned all care of his possessions, as of himself, wherefore the most part of the houses were become common good and strangers used them, whenas they happened upon them, like as the very owner might have done; and with all this bestial preoccupation, they still shunned the sick to the best of their power.

In this sore affliction and misery of our city, the reverend authority of the laws, both human and divine, was all in a manner dissolved and fallen into decay, for the ministers and executors thereof, who, like other men, were all either dead or sick or else left so destitute of followers that they were unable to exercise any office, wherefore every one had license to do whatsoever pleased him. Many others held a middle course between the two aforesaid, not straitening themselves so exactly in the matter of diet as the first neither allowing themselves such license in drinking and other debauchery as the second, but using things in sufficiency, according to their appetites; nor did they seclude themselves, but went about, carrying in their hands, some flowers, some odoriferous herbs and other some divers kinds of spiceries, which they set often to their noses, accounting it an excellent thing to fortify the brain with such odours, more by token that the air seemed all heavy and attainted with the stench of the dead bodies and that of the sick and of the remedies used.

Some were of a more barbarous, though, peradventure, a surer way of thinking, avouching that there was no remedy against pestilences better than—no, nor any so good as—to flee before them; wherefore, moved by this reasoning and recking of nought but themselves, very many, both men and women, abandoned their own city, their own houses and homes, their kinsfolk and possessions, and sought the country seats of others, or, at the least, their own, as if the wrath of God, being moved to punish the iniquity of mankind, would not proceed to do so wheresoever they might be, but would content itself with afflicting those only who were found within the walls of their city, or as if they were persuaded that no person was to remain therein and that its last hour was come. And albeit these, who opined thus variously, died not all, yet neither did they all escape; nay, many of each way of thinking and in every place sickened of the plague and languished on all sides, well nigh abandoned, having themselves, what while they were whole, set the example to those who abode in health.

Indeed, leaving be that townsman avoided townsman and that well nigh no neighbour took thought unto other and that kinsfolk seldom or never visited one another and held no converse together save from afar, this tribulation had stricken such terror to the hearts of all, men and women alike, that brother forsook brother, uncle nephew and sister brother and oftentimes wife husband; nay (what is yet more extraordinary and well nigh incredible) fathers and mothers refused to visit or tend their very children, as they had not been theirs. By reason whereof there remained unto those (and the number of them, both males and females, was incalculable) who fell sick, none other succour than that which they

owed either to the charity of friends (and of these there were few) or the greed of servants, who tended them, allured by high and extravagant wage; albeit, for all this, these latter were not grown many, and those men and women of mean understanding and for the most part unused to such offices, who served for well nigh nought but to reach things called for by the sick or to note when they died; and in the doing of these services many of them perished with their gain.

Of this abandonment of the sick by neighbours, kinsfolk and friends and of the scarcity of servants arose an usage before well nigh unheard, to wit, that no woman, how fair or lovesome or well-born soever she might be, once fallen sick, recked aught of having a man to tend her, whatever he might be, or young or old, and without any shame discovered to him every part of her body, no otherwise than she would have done to a woman, so but the necessity of her sickness required it; the which belike, in those who recovered, was the occasion of lesser modesty in time to come. Moreover, there ensued of this abandonment the death of many who peradventure, had they been succoured, would have escaped alive; wherefore, as well for the lack of the opportune services which the sick availed not to have as for the virulence of the plague, such was the multitude of those who died in the city by day and by night that it was an astonishment to hear tell thereof, much more to see it; and thence, as it were of necessity, there sprang up among those who abode alive things contrary to the pristine manners of the townsfolk.

It was then (even as we yet see it used) a custom that the kinswomen and she-neighbours of the dead should assemble in his house and there condole with those who more nearly pertained unto him, whilst his neighbours and many other citizens foregathered with his next of kin before his house, whither, according to the dead man's quality, came the clergy, and he with funeral pomp of chants and candles was borne on the shoulders of his peers to the church chosen by himself before his death; which usages, after the virulence of the plague began to increase, were either altogether or for the most part laid aside, and other and strange customs sprang up in their stead. For that, not only did folk die without having a multitude of women about them, but many there were who departed this life without witness and few indeed were they to whom the pious complaints and bitter tears of their kinsfolk were vouchsafed; nay, in lieu of these things there obtained, for the most part, laughter and jests and gibes and feasting and merrymaking in company; which usance women, laying aside womanly pitifulness, had right well learned for their own safety.

Few, again, were they whose bodies were accompanied to the church by more than half a score or a dozen of their neighbours, and of these no worshipful and illustrious citizens, but a sort of blood-suckers, sprung from the dregs of the people, who styled themselves pickmen and did such offices for hire, shouldered the bier and bore it with hurried steps, not to that church which the dead man had chosen before his death, but most times to the nearest, behind five or six priests, with little light and whiles none at all, which latter, with the aid of the said pickmen, thrust him into what grave soever they first found unoccupied, without troubling themselves with too long or too formal a service.

The condition of the common people (and belike, in great part, of the middle class also) was yet more pitiable to behold, for that these, for the most part retained by hope or poverty in their houses and abiding in their own quarters, sickened by the thousand daily and being altogether untended and unsuccoured, died well nigh all without recourse. Many breathed their last in the open street, whilst other many, for all they died in their houses, made it known to the neighbours that they were dead rather by the stench of their rotting bodies than otherwise; and of these and others who died all about the whole city was full. For the most part one same usance was observed by the neighbours, moved more by fear lest the corruption of the dead bodies should imperil themselves than by any charity they had for the departed; to wit, that either with their own hands or with the aid of certain bearers, whenas

they might have any, they brought the bodies of those who had died forth of their houses and laid them before their doors, where, especially in the morning, those who went about might see corpses without number; then they fetched biers and some, in default thereof, they laid upon some board or other. Nor was it only one bier that carried two or three corpses, nor did this happen but once; nay, many might have been counted which contained husband and wife, two or three brothers, father and son or the like. And an infinite number of times it befell that, two priests going with one cross for some one, three or four biers, borne by bearers, ranged themselves behind the latter, and whereas the priests thought to have but one dead man to bury, they had six or eight, and whiles more. Nor therefore were the dead honoured with aught of tears or candles or funeral train; nay, the thing was come to such a pass that folk recked no more of men that died than nowadays they would of goats; whereby it very manifestly appeared that that which the natural course of things had not availed, by dint of small and infrequent harms, to teach the wise to endure with patience, the very greatness of their ills had brought even the simple to expect and make no account of. The consecrated ground sufficing not to the burial of the vast multitude of corpses aforesaid, which daily and well nigh hourly came carried in crowds to every church,—especially if it were sought to give each his own place, according to ancient usance,—there were made throughout the churchyards, after every other part was full, vast trenches, wherein those who came after were laid by the hundred and being heaped up therein by layers, as goods are stowed aboard ship, were covered with a little earth, till such time as they reached the top of the trench.

Moreover,—not to go longer searching out and recalling every particular of our past miseries, as they befell throughout the city,—I say that, whilst so sinister a time prevailed in the latter, on no wise therefor was the surrounding country spared, wherein, (letting be the castles, which in their littleness were like unto the city) throughout the scattered villages and in the fields, the poor and miserable husbandmen and their families, without succour of physician or aid of servitor, died, not like men, but well nigh like beasts, by the ways or in their tillages or about the houses, indifferently by day and night. By reason whereof, growing lax like the townsfolk in their manners and customs, they recked not of any thing or business of theirs; nay, all, as if they looked for death that very day, studied with all their wit, not to help to maturity the future produce of their cattle and their fields and the fruits of their own past toils, but to consume those which were ready to hand. Thus it came to pass that the oxen, the asses, the sheep, the goats, the swine, the fowls, nay, the very dogs, so faithful to mankind, being driven forth of their own houses, went straying at their pleasure about the fields, where the very corn was abandoned, without being cut, much less gathered in; and many, well nigh like reasonable creatures, after grazing all day, returned at night, gluttled, to their houses, without the constraint of any herdsman.

To leave the country and return to the city, what more can be said save that such and so great was the cruelty of heaven (and in part, peradventure, that of men) that, between March and the following July, what with the virulence of that pestiferous sickness and the number of sick folk ill tended or forsaken in their need, through the fearfulness of those who were whole, it is believed for certain that upward of an hundred thousand human beings perished within the walls of the city of Florence, which, peradventure, before the advent of that death-dealing calamity, had not been accounted to hold so many? Alas, how many great palaces, how many goodly houses, how many noble mansions, once full of families, of lords and of ladies, abode empty even to the meanest servant! How many memorable families, how many ample heritages, how many famous fortunes were seen to remain without lawful heir! How many valiant men, how many fair ladies, how many sprightly youths, whom, not others only, but Galen, Hippocrates or Æsculapius themselves would have judged most hale, breakfasted in the morning with their kinsfolk, comrades and friends and that same night supped with their ancestors in the other world!

I am myself weary of going wandering so long among such miseries; wherefore, purposing

henceforth to leave such part thereof as I can fitly, I say that,—our city being at this pass, well nigh void of inhabitants,—it chanced (as I afterward heard from a person worthy of credit) that there foregathered in the venerable church of Santa Maria Novella, one Tuesday morning when there was well nigh none else there, seven young ladies, all knit one to another by friendship or neighbourhood or kinship, who had heard divine service in mourning attire, as sorted with such a season. Not one of them had passed her eight-and-twentieth year nor was less than eighteen years old, and each was discreet and of noble blood, fair of favour and well-mannered and full of honest sprightliness. The names of these ladies I would in proper terms set out, did not just cause forbid me, to wit, that I would not have it possible that, in time to come, any of them should take shame by reason of the things hereinafter related as being told or hearkened by them, the laws of disport being nowadays somewhat straitened, which at that time, for the reasons above shown, were of the largest, not only for persons of their years, but for those of a much ripper age; nor yet would I give occasion to the envious, who are still ready to carp at every praiseworthy life, on anywise to disparage the fair fame of these honourable ladies with unseemly talk. Wherefore, so that which each saith may hereafterward be apprehended without confusion, I purpose to denominate them by names altogether or in part sorting with each one's quality. The first of them and her of ripest age I shall call Pampinea, the second Fiammetta, the third Filomena and the fourth Emilia. To the fifth we will give the name of Lauretta, to the sixth that of Neifile and the last, not without cause, we will style Elisa. These, then, not drawn of any set purpose, but foregathering by chance in a corner of the church, having seated themselves in a ring, after divers sighs, let be the saying of paternosters and fell to devising with one another many and various things of the nature of the time. After awhile, the others being silent, Pampinea proceeded to speak thus:

"Dear my ladies, you may, like myself, have many times heard that whoso honestly useth his right doth no one wrong; and it is the natural right of every one who is born here below to succour, keep and defend his own life as best he may, and in so far is this allowed that it hath happened whiles that, for the preservation thereof, men have been slain without any fault. If this much be conceded of the laws, which have in view the well-being of all mortals, how much more is it lawful for us and whatsoever other, without offence unto any, to take such means as we may for the preservation of our lives? As often as I consider our fashions of this morning and those of many other mornings past and bethink me what and what manner discourses are ours, I feel, and you likewise must feel, that each of us is in fear for herself. Nor do I anywise wonder at this; but I wonder exceedingly, considering that we all have a woman's wit, that we take no steps to provide ourselves against that which each of us justly feareth. We abide here, to my seeming, no otherwise than as if we would or should be witness of how many dead bodies are brought hither for burial or to hearken if the friars of the place, whose number is come well nigh to nought, chant their offices at the due hours or by our apparel to show forth unto whosoever appeareth here the nature and extent of our distresses. If we depart hence, we either see dead bodies or sick persons carried about or those, whom for their misdeeds the authority of the public laws whilere condemned to exile, overrun the whole place with unseemly excesses, as if scoffing at the laws, for that they know the executors thereof to be either dead or sick; whilst the dregs of our city, fattened with our blood, style themselves pickmen and ruffle it everywhere in mockery of us, riding and running all about and flouting us with our distresses in ribald songs. We hear nothing here but 'Such an one is dead' or 'Such an one is at the point of death'; and were there any to make them, we should hear dolorous lamentations on all sides. And if we return to our houses, I know not if it is with you as with me, but, for my part, when I find none left therein of a great household, save my serving-maid, I wax fearful and feel every hair of my body stand on end; and wherever I go or abide about the house, meseemeth I see the shades of those who are departed and who wear not those countenances that I was used to see, but terrify me with a horrid aspect, I know not whence newly come to them.

By reason of these things I feel myself alike ill at ease here and abroad and at home, more by

token that meseemeth none, who hath, as we have, the power and whither to go, is left here, other than ourselves; or if any such there be, I have many a time both heard and perceived that, without making any distinction between things lawful and unlawful, so but appetite move them, whether alone or in company, both day and night, they do that which affordeth them most delight. Nor is it the laity alone who do thus; nay, even those who are shut in the monasteries, persuading themselves that what befitteth and is lawful to others alike sortable and unforbidden unto them, have broken the laws of obedience and giving themselves to carnal delights, thinking thus to escape, are grown lewd and dissolute. If thus, then, it be, as is manifestly to be seen, what do we here? What look we for? What dream we? Why are we more sluggish and slower to provide for our safety than all the rest of the townsfolk? Deem we ourselves of less price than others, or do we hold our life to be bounden in our bodies with a stronger chain than is theirs and that therefore we need reckon nothing of aught that hath power to harm it? We err, we are deceived; what folly is ours, if we think thus! As often as we choose to call to mind the number and quality of the youths and ladies overborne of this cruel pestilence, we may see a most manifest proof thereof.

Wherefore, in order that we may not, through wilfulness or nonchalance, fall into that wherefrom we may, peradventure, an we but will, by some means or other escape, I know not if it seem to you as it doth to me, but methinketh it were excellently well done that we, such as we are, depart this city, as many have done before us, and eschewing, as we would death, the dishonourable example of others, betake ourselves quietly to our places in the country, whereof each of us hath great plenty, and there take such diversion, such delight and such pleasure as we may, without anywise overpassing the bounds of reason. There may we hear the small birds sing, there may we see the hills and plains clad all in green and the fields full of corn wave even as doth the sea; there may we see trees, a thousand sorts, and there is the face of heaven more open to view, the which, angered against us though it be, nevertheless denieth not unto us its eternal beauties, far goodlier to look upon than the empty walls of our city. Moreover, there is the air far fresher and there at this season is more plenty of that which behoveth unto life and less is the sum of annoys, for that, albeit the husbandmen die there, even as do the townsfolk here, the displeasure is there the less, insomuch as houses and inhabitants are rarer than in the city.

Here, on the other hand, if I deem aright, we abandon no one; nay, we may far rather say with truth that we ourselves are abandoned, seeing that our kinsfolk, either dying or fleeing from death, have left us alone in this great tribulation, as it were we pertained not unto them. No blame can therefore befall the ensuing of this counsel; nay, dolour and chagrin and belike death may betide us, an we ensue it not. Wherefore, an it please you, methinketh we should do well to take our maids and letting follow after us with the necessary gear, sojourn to-day in this place and to-morrow in that, taking such pleasure and diversion as the season may afford, and on this wise abide till such time (an we be not earlier overtaken of death) as we shall see what issue Heaven reserveth unto these things. And I would remind you that it is no more forbidden unto us honourably to depart than it is unto many others of our sex to abide in dishonour."

The other ladies, having hearkened to Pampinea, not only commended her counsel, but, eager to follow it, had already begun to devise more particularly among themselves of the manner, as if, arising from their session there, they were to set off out of hand. But Filomena, who was exceeding discreet, said, "Ladies, albeit that which Pampinea allegeth is excellently well said, yet is there no occasion for running, as meseemeth you would do. Remember that we are all women and none of us is child enough not to know how reasonable women are among themselves and how, without some man's guidance, they know how to order themselves. We are fickle, wilful, suspicious, faint-hearted and timorous, for which reasons I misdoubt me sore, an we take not some other guidance than our own, that our company

will be far too soon dissolved and with less honour to ourselves than were seemly; wherefore we should do well to provide ourselves, ere we begin."

"Verily," answered Elisa, "men are the head of women, and without their ordinance seldom cometh any emprise of ours to good end; but how may we come by these men? There is none of us but knoweth that of her kinsmen the most part are dead and those who abide alive are all gone fleeing that which we seek to flee, in divers companies, some here and some there, without our knowing where, and to invite strangers would not be seemly, seeing that, if we would endeavour after our welfare, it behoveth us find a means of so ordering ourselves that, wherever we go for diversion and repose, scandal nor annoy may ensue thereof."

Whilst such discourse was toward between the ladies, behold, there entered the church three young men,—yet not so young that the age of the youngest of them was less than five-and-twenty years,—in whom neither the perversity of the time nor loss of friends and kinsfolk, no, nor fear for themselves had availed to cool, much less to quench, the fire of love. Of these one was called Pamfilo, another Filostrato and the third Dioneo, all very agreeable and well-bred, and they went seeking, for their supreme solace, in such a perturbation of things, to see their mistresses, who, as it chanced, were all three among the seven aforesaid; whilst certain of the other ladies were near kinswomen of one or other of the young men.

No sooner had their eyes fallen on the ladies than they were themselves espied of them; whereupon quoth Pampinea, smiling, "See, fortune is favourable to our beginnings and hath thrown in our way young men of worth and discretion, who will gladly be to us both guides and servitors, and we disdain not to accept of them in that capacity." But Neifile, whose face was grown all vermeil for shamefastness, for that it was she who was beloved of one of the young men, said, "For God's sake, Pampinea, look what thou sayest! I acknowledge most frankly that there can be nought but all good said of which one soever of them and I hold them sufficient unto a much greater thing than this, even as I opine that they would bear, not only ourselves, but far fairer and nobler dames than we, good and honourable company. But, for that it is a very manifest thing that they are enamoured of certain of us who are here, I fear lest, without our fault or theirs, scandal and blame ensue thereof, if we carry them with us." Quoth Filomena, "That skilleth nought; so but I live honestly and conscience prick me not of aught, let who will speak to the contrary; God and the truth will take up arms for me. Wherefore, if they be disposed to come, verily we may say with Pampinea that fortune is favourable to our going."

The other ladies, hearing her speak thus absolutely, not only held their peace, but all with one accord agreed that the young men should be called and acquainted with their project and bidden to be pleased bear them company in their expedition. Accordingly, without more words, Pampinea, who was knit by kinship to one of them, rising to her feet, made for the three young men, who stood fast, looking upon them, and saluting them with a cheerful countenance, discovered to them their intent and prayed them, on behalf of herself and her companions, that they would be pleased to bear them company in a pure and brotherly spirit. The young men at the first thought themselves bantered, but, seeing that the lady spoke in good earnest, they made answer joyfully that they were ready, and without losing time about the matter, forthright took order for that which they had to do against departure.

On the following morning, Wednesday to wit, towards break of day, having let orderly make ready all things needful and despatched them in advance whereas they purposed to go, the ladies, with certain of their waiting-women, and the three young men, with as many of their serving-men, departing Florence, set out upon their way; nor had they gone more than two short miles from the city, when they came to the place fore-appointed of them, which was situate on a little hill, somewhat withdrawn on

every side from the high way and full of various shrubs and plants, all green of leafage and pleasant to behold. On the summit of this hill was a palace, with a goodly and great courtyard in its midst and galleries [loggias] and saloons and bedchambers, each in itself most fair and adorned and notable with jocund paintings, with lawns and grassplots round about and wonder-goodly gardens and wells of very cold water and cellars full of wines of price, things more apt unto curious drinkers than unto sober and modest ladies. The new comers, to their no little pleasure, found the place all swept and the beds made in the chambers and every thing full of such flowers as might be had at that season and strewn with rushes.

As soon as they had seated themselves, Dioneo, who was the merriest springald in the world and full of quips and cranks, said, "Ladies, your wit, rather than our foresight, hath guided us hither, and I know not what you purpose to do with your cares; as for my own, I left them within the city gates, whenas I issued thence with you awhile ago; wherefore, do you either address yourselves to make merry and laugh and sing together with me (in so far, I mean, as pertaineth to your dignity) or give me leave to go back for my cares and abide in the afflicted city." Whereto Pampinea, no otherwise than as if in like manner she had banished all her own cares, answered blithely, "Dioneo, thou sayst well; it behoveth us live merrily, nor hath any other occasion caused us flee from yonder miseries. But, for that things which are without measure may not long endure, I, who began the discourse wherethrough this so goodly company came to be made, taking thought for the continuance of our gladness, hold it of necessity that we appoint some one to be principal among us, whom we may honour and obey as chief and whose especial care it shall be to dispose us to live joyously. And in order that each in turn may prove the burden of solicitude, together with the pleasure of headship; and that, the chief being thus drawn, in turn, from one and the other sex, there may be no cause for jealousy, as might happen, were any excluded from the sovranty, I say that unto each be attributed the burden and the honour for one day. Let who is to be our first chief be at the election of us all. For who shall follow, be it he or she whom it shall please the governor of the day to appoint, whenas the hour of vespers draweth near, and let each in turn, at his or her discretion, order and dispose of the place and manner wherein we are to live, for such time as his or her seignory shall endure."

Pampinea's words pleased mightily, and with one voice they elected her chief of the first day; whereupon Filomena, running nimbly to a laurel-tree—for that she had many a time heard speak of the honour due to the leaves of this plant and how worship-worth they made whoso was deservedly crowned withal—and plucking divers sprays therefrom, made her thereof a goodly and honourable wreath, which, being set upon her head, was thenceforth, what while their company lasted, a manifest sign unto every other of the royal office and seignory.

Pampinea, being made queen, commanded that every one should be silent; then, calling the serving-men of the three young gentlemen and her own and the other ladies' women, who were four in number, before herself and all being silent, she spoke thus: "In order that I may set you a first example, by which, proceeding from good to better, our company may live and last in order and pleasance and without reproach so long as it is agreeable to us, I constitute, firstly, Parmeno, Dioneo's servant, my seneschal and commit unto him the care and ordinance of all our household and that which pertaineth to the service of the saloon. Sirisco, Pamfilo's servant, I will shall be our purveyor and treasurer and ensue the commandments of Parmeno. Tindaro shall look to the service of Filostrato and the other two gentlemen in their bed chambers, what time the others, being occupied about their respective offices, cannot attend thereto. Misia, my woman, and Filomena's Licisca shall still abide in the kitchen and there diligently prepare such viands as shall be appointed them of Parmeno. Lauretta's Chimera and Fiammetta's Stratilia it is our pleasure shall occupy themselves with the ordinance of the ladies' chambers and the cleanliness of the places where we shall abide; and we will and command all and

several, as they hold our favour dear, to have a care that, whithersoever they go or whencesoever they return and whatsoever they hear or see, they bring us from without no news other than joyous." These orders summarily given and commended of all, Pampinea, rising blithely to her feet, said, "Here be gardens, here be meadows, here be store of other delectable places, wherein let each go a-pleasuring at will; and when tierce [Nine o'clock a.m.] soundeth, let all be here, so we may eat in the cool."

The merry company, being thus dismissed by the new queen, went straying with slow steps, young men and fair ladies together, about a garden, devising blithely and diverting themselves with weaving goodly garlands of various leaves and carolling amorously. After they had abidden there such time as had been appointed them of the queen, they returned to the house, where they found that Parmeno had made a diligent beginning with his office, for that, entering a saloon on the ground floor, they saw there the tables laid with the whitest of cloths and beakers that seemed of silver and everything covered with the flowers of the broom; whereupon, having washed their hands, they all, by command of the queen, seated themselves according to Parmeno's ordinance. Then came viands delicately drest and choicest wines were proffered and the three serving-men, without more, quietly tended the tables. All, being gladdened by these things, for that they were fair and orderly done, ate joyously and with store of merry talk, and the tables being cleared away [the table of Boccaccio's time was a mere board upon trestles, which when not in actual use, was stowed away, for room's sake, against the wall.], the queen bade bring instruments of music, for that all the ladies knew how to dance, as also the young men, and some of them could both play and sing excellent well. Accordingly, by her commandment, Dioneo took a lute and Fiammetta a viol and began softly to sound a dance; whereupon the queen and the other ladies, together with the other two young men, having sent the serving-men to eat, struck up a round and began with a slow pace to dance a brawl; which ended, they fell to singing quaint and merry ditties. On this wise they abode till it seemed to the queen time to go to sleep, and she accordingly dismissed them all; whereupon the young men retired to their chambers, which were withdrawn from the ladies' lodging, and finding them with the beds well made and as full of flowers as the saloon, put off their clothes and betook themselves to rest, whilst the ladies, on their part, did likewise.

None [Three o'clock p.m.] had not long sounded when the queen, arising, made all the other ladies arise, and on like wise the three young men, alleging overmuch sleep to be harmful by day; and so they betook themselves to a little meadow, where the grass grew green and high nor there had the sun power on any side. There, feeling the waftings of a gentle breeze, they all, as their queen willed it, seated themselves in a ring on the green grass; while she bespoke them thus, "As ye see, the sun is high and the heat great, nor is aught heard save the crickets yonder among the olives; wherefore it were doubtless folly to go anywhither at this present. Here is the sojourn fair and cool, and here, as you see, are chess and tables [backgammon], and each can divert himself as is most to his mind. But, and my counsel be followed in this, we shall pass away this sultry part of the day, not in gaming,—wherein the mind of one of the players must of necessity be troubled, without any great pleasure of the other or of those who look on,—but in telling stories, which, one telling, may afford diversion to all the company who hearken; nor shall we have made an end of telling each his story but the sun will have declined and the heat be abated, and we can then go a-pleasuring whereas it may be most agreeable to us. Wherefore, if this that I say please you, (for I am disposed to follow your pleasure therein,) let us do it; and if it please you not, let each until the hour of vespers do what most liketh him." Ladies and men alike all approved the story-telling, whereupon, "Then," said the queen, "since this pleaseth you, I will that this first day each be free to tell of such matters as are most to his liking." Then, turning to Pamfilo, who sat on her right hand, she smilingly bade him give beginning to the story-telling with one of his; and he, hearing the commandment, forthright began thus, whilst all gave ear to him ...

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On the next page

Plague column (one of 3 in Naples). Spire of San Domenico (1737), square of San Domenico Maggiore, Naples, Italy with statue of Saint Dominic, commemorating the plague of 1656.



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